Kunj Sonker

Email: iec2024110@iiita.ac.in Phone: +91 7800889088 LinkedIn: linkedin.com/in/kunjsonker
GitHub: github.com/kunjsonker

Education

Indian Institute of Information Technology Allahabad 2024 – 2028(Expected)
B.Tech in Electronics and Communication Engineering **CGPA: 7.8/10** (till 4th Sem)

Technical Skills

Languages:	Python, C++, C, Java, JavaScript (ES6+), TypeScript, HTML, CSS
Frameworks:	Node.js, Express.js, React.js, Next.js, Tailwind CSS, Framer Motion, Streamlit
Data Science:	scikit-learn, pandas, NumPy, DBSCAN, PyDeck, Geospatial Analytics, Haversine
Databases/Cloud:	MongoDB Atlas, PostgreSQL, MySQL, Firebase, Cloudinary, Streamlit Community Cloud
Tools:	Git, GitHub, Postman, JWT, Bcrypt, Multer, Axios, REST APIs, Vercel, Render
Soft Skills:	Analytical Thinking, Problem Solving, Self-Motivated, Detail-Oriented, Collaborative, Adaptable

Projects

ClearWay – Traffic Enforcement Intelligence System *GitHub* | *Live*

- **End-to-End ML Pipeline:** Built and deployed ClearWay — an end-to-end traffic enforcement intelligence system processing **300,000+ real parking violation records** spanning **12 months** of historical data using **DBSCAN spatial clustering** with Haversine distance, detecting **active hotspots** from live Bengaluru data.
- **Custom Scoring Model:** Designed a **Traffic Impact Score (TIS)** combining vehicle lane-blocking weight and obstruction duration to rank **enforcement zones** across **5+ violation categories**, shifting dispatch from random patrol to data-backed prioritization.
- **Geospatial Dashboard:** Built and deployed a live **3D geospatial dashboard** using **Streamlit** and **PyDeck** with real-time filtering across **200 km²** of Bengaluru's road network, hourly trend analytics and one-click CSV export — publicly accessible on Streamlit Community Cloud.

Appointo – Full Stack Health-Tech Platform *GitHub* | *Live*

- **Multi-Role System:** Engineered **3** independent panels for Patients, Doctors and Administrators, streamlining patient-doctor interactions with distinct dashboards, role-based access control and dedicated workflows deployed on cloud infrastructure.
- **Distributed Architecture:** Built a MERN backend using **MVC architecture** with **10+ RESTful API endpoints** to manage high-concurrency appointment scheduling across multiple user roles and services.
- **Performance:** Achieved average API latency of **142ms** — **43% below** the industry-standard 250ms target — through optimized **MongoDB Atlas** indexing and JSON payload management (avg. 2.1KB).
- **Security:** Implemented **stateless JWT authentication** and **Bcrypt** hashing with custom middleware enforcing role-based access control (RBAC) across all **3** panels.
- **File Handling:** Integrated **Multer** for multipart file uploads and **Cloudinary** for scalable cloud-based storage and delivery of **70+** media assets.

Next.js Performance Portfolio *GitHub* | *Live*

- **Modern Web Stack:** Built using **Next.js** and **TypeScript**, achieving **96/100 Google Lighthouse** performance score and **100/100 SEO** score.
- **Optimization:** Reduced **LCP to 0.8s** — **68% faster** than the 2.5s Core Web Vitals benchmark — via Server-Side Rendering (SSR), Next/Image optimization and priority asset loading.
- **GPU Rendering:** Engineered a custom particle interaction system using **HTML5 Canvas API** and **Framer Motion**, maintaining stable **60 FPS** through GPU-accelerated rendering.

Achievements

- **Competitive Programming:** Solved **400+ problems** on LeetCode, CodeChef, and Codeforces; rated **Pupil** on Codeforces and **3-Star** on CodeChef; secured global ranks **90** (Div 3), **488** (Div 3), **671** (Div 2) among 65,000+ participants.
- **Amazon HackOn'26:** Ranked in the **Top 75** coders nationwide in Amazon's premier coding competition.
- **Flipkart Gridlock' 26:** Qualified as a **Semifinalist** in Flipkart's flagship national engineering challenge.
- **Amazon MLSS'26:** Qualified for Amazon's ML Summer School (MLSS) among **130,000+** candidates nationwide.

Relevant Coursework

Data Structures & Algorithms (DSA)	Microprocessor Interface	Network Theory
Problem Solving (C/C++)	Linear Algebra	